

9 Comparative Statics and Boundary Cases

This section distinguishes four operations that cannot be interchanged:

1. a derivative with respect to a continuous primitive at a fixed candidate CMO price p_m ;
2. a derivative with respect to a continuous primitive after the CMO market clears at p_m^* ;
3. a finite comparison between the binary regimes $M = 0$ and $M = 1$;
4. a limiting-case calculation.

There is no derivative with respect to binary M . Before each economic interpretation, the mathematical margin is named as commercial return, commercialization-route choice, project advancement, the CMO price, or realized product output. Upstream research and patent generation are not endogenous margins of the baseline.

9.1 Commercial pricing and return

Object and held-fixed variables. The differentiated objects are the optimized product price, operating profit, and commercial present value. Project value q , marginal manufacturing cost c , and the demand and discount primitives are held fixed except for the single argument named in each derivative. These are conditional product-market calculations; neither M nor p_m^* is varied.

The pricing solution and operating profit derived above can be written as

$$p^*(c) = \frac{\varepsilon}{\varepsilon - 1}c, \quad \pi(q, c) = \frac{Aq}{\varepsilon} \left(\frac{\varepsilon}{\varepsilon - 1}c \right)^{1-\varepsilon}.$$

Direct differentiation gives

$$\frac{\partial p^*}{\partial c} = \frac{\varepsilon}{\varepsilon - 1} > 1, \quad \frac{\partial \pi}{\partial q} = \frac{\pi}{q} > 0, \quad \frac{\partial \pi}{\partial c} = (1 - \varepsilon)\frac{\pi}{c} < 0. \quad (1)$$

Since $R(q, c) = \pi(q, c)/(1 - \beta\varphi)$,

$$R_q = \frac{R}{q} > 0, \quad R_c = (1 - \varepsilon)\frac{R}{c} < 0, \quad (2)$$

and, holding π fixed,

$$R_\beta = \frac{\varphi\pi}{(1 - \beta\varphi)^2} \geq 0, \quad R_\varphi = \frac{\beta\pi}{(1 - \beta\varphi)^2} > 0.$$

The first inequality is an equality when $\varphi = 0$.

Sufficient conditions, dimensions, and zero effects. The signs use $A, q, c > 0$, $\varepsilon > 1$, $\beta \in (0, 1)$, and $\varphi \in [0, 1)$. The price derivative is dimensionless; R_q and R_c have currency per unit of their respective arguments. A change in β has no continuation-value effect when $\varphi = 0$. None of these derivatives is a policy effect unless a specified route technology changes the cost argument supplied to R .

Economic interpretation and empirical prediction. This is a commercial-return margin. Higher manufacturing cost is passed through to the product price and lowers optimized commercial value, while a higher project-value shifter raises it. The empirical prediction is conditional cost pass-through and lower commercial return for more costly manufacturing, not a claim about patenting or upstream research.

9.2 Commercialization-route choice at fixed CMO price

Object and held-fixed variables. The object is the internal-minus-entrusted value gap and its conditional capability cutoff. Hold (q, m) , all technologies and outside options fixed. For capability derivatives also hold finite (τ_E, p_m) fixed. For each implicit cutoff derivative, vary only the continuous argument displayed.

On the interior of the internally feasible domain,

$$\Delta_{IE,k} = s(q)R_c(q, c_I)c_{I,k} - F_{I,k} > 0. \quad (3)$$

The operating-cost component is weakly positive and may be zero when $s(q) = 0$; the setup-cost component is strictly positive. Under the cell-specific crossing condition, the implicit-function theorem gives

$$\left. \frac{\partial k^*}{\partial \tau_E} \right|_{q,m,p_m} = -\frac{1}{\Delta_{IE,k}(k^*)} < 0, \quad \left. \frac{\partial k^*}{\partial p_m} \right|_{q,m,\tau_E} = -\frac{b(m)}{\Delta_{IE,k}(k^*)} < 0. \quad (4)$$

Both are continuous-argument derivatives inside the finite-wedge domain. They are not derivatives with respect to M , and the second is not an equilibrium-price effect.

Binary policy comparison. Let $W_i^0 = \max\{W_i^I, W_i^T, 0\}$. At any fixed finite p_m ,

$$W_i(q, m; 1, p_m) - W_i(q, m; 0, p_m) = [W_i^E(q, m; 1, p_m) - W_i^0(q, m)]_+ \geq 0. \quad (5)$$

This finite comparison adds an option without changing the old route values. It is strict only on $\mathcal{C}_i(p_m)$.

Sufficient conditions and zero effects. The cutoff signs require internal feasibility, the maintained technology signs, finite (τ_E, p_m) , strict $\Delta_{IE,k} > 0$, and endpoint crossing. If crossing fails, no interior cutoff is asserted. If T or A dominates, the I/E cutoff does not determine the realized route. If $W_i^E \leq W_i^0$, the finite policy gain is zero.

Economic interpretation and empirical prediction. This is a commercialization-route margin. Better internal manufacturing capability makes internal production relatively more attractive; a lower finite entrusted-route burden expands the capability region using E , while a higher fixed CMO price contracts it. The empirical prediction is sorting of holder–manufacturer organization by predetermined manufacturing capability and project requirement, conditional on outside options.

9.3 Project advancement at fixed CMO price

Object and held-fixed variables. The object is the unique common project-advancement intensity

$$x_i^*(M, p_m) = \left[\frac{\beta a_i \Omega_i(M, p_m)}{\kappa} \right]^{1/\nu}.$$

Each partial derivative below holds M, p_m, ν and all unlisted primitives fixed. In particular, route-value objects are held fixed when a_i or κ is varied. The formulas apply at $\Omega_i > 0$; the $\Omega_i = 0$ corner is handled separately.

Differentiation of the advancement FOC gives

$$\frac{\partial x_i^*}{\partial \Omega_i} = \frac{x_i^*}{\nu \Omega_i} > 0, \quad \frac{\partial x_i^*}{\partial a_i} = \frac{x_i^*}{\nu a_i} > 0, \quad \frac{\partial x_i^*}{\partial \kappa} = -\frac{x_i^*}{\nu \kappa} < 0. \quad (6)$$

The corresponding elasticities are $1/\nu, 1/\nu, -1/\nu$. Because the optimized project value obeys the deterministic envelope, $\Omega_{i,p_m} = -\chi_i^E$ almost everywhere, so

$$\left. \frac{\partial x_i^*}{\partial p_m} \right|_M = -\frac{x_i^*}{\nu \Omega_i} \chi_i^E \leq 0. \quad (7)$$

This is a fixed-candidate-price derivative and already includes both the intensive value effect and deterministic route-envelope effect inside Ω_i .

For binary M , define at a common support price

$$\Delta \Omega_i(p_m) = \int [W_i^E(q, m; 1, p_m) - W_i^0(q, m)]_+ dF.$$

The correct policy calculation is the finite difference

$$\Delta x_i(p_m) = \left(\frac{\beta a_i}{\kappa} \right)^{1/\nu} \left[(\Omega_i^0 + \Delta \Omega_i(p_m))^{1/\nu} - (\Omega_i^0)^{1/\nu} \right] \geq 0. \quad (8)$$

It is strict if and only if $\Delta \Omega_i(p_m) > 0$.

Sufficient conditions and zero effects. The partials require $\Omega_i > 0$. At the zero-value corner, $x_i^* = 0$; global monotonicity follows directly from the closed form. Higher a_i scales a positive response but cannot create one when the entrusted surplus is zero. The weak price effect is zero when no project uses E , so $\chi_i^E = 0$. The manufacturing-capability ordering of the level response additionally uses $\nu \geq 1$; no such sign is claimed for $0 < \nu < 1$ without another bound.

Economic interpretation and empirical prediction. This is development-stage project advancement, not upstream research. Expected commercialization value raises the return to advancing viable projects; higher advancement productivity amplifies a positive value gain, whereas a steeper cost scale reduces the chosen intensity. The prediction is a larger project-advancement response among high- a_i developers for which entrusted manufacturing adds value, especially when internal manufacturing capability is weak under the stated curvature condition. No patent-application response follows mechanically.

9.4 CMO supply, demand, and continuous equilibrium derivatives

Supplier object and held-fixed variables. For a supplier with $p_m > 0$, vary only candidate price or efficiency while holding the other argument and the cost function fixed. Implicit differentiation of $p_m = \Psi_s(s_j^*; z_j)$ gives

$$\frac{\partial s_j^*}{\partial p_m} = \frac{1}{\Psi_{ss}} > 0, \quad \frac{\partial s_j^*}{\partial z_j} = -\frac{\Psi_{sz}}{\Psi_{ss}} > 0. \quad (9)$$

These are capacity-supply derivatives; policy does not shift Ψ or z_j .

Study-demand object. For this local formula, assume differentiability and an integrable envelope for local difference quotients, permitting differentiation under the developer integral. Then

$$\frac{\partial D_m^{\text{MAH}}}{\partial p_m} = \int a_i \left[\frac{\partial x_i^*}{\partial p_m} \chi_i^E + x_i^* \frac{\partial \chi_i^E}{\partial p_m} \right] dH(a, k) \leq 0. \quad (10)$$

The first term is the project-value/advancement response; the second is the commercialization-route response. Where a classical derivative does not exist, the global weak decrease follows from the product of nonnegative monotone responses. The no-tie and integrability assumptions ensure continuity, not necessarily classical differentiability.

Continuous equilibrium perturbation. Consider an infinitesimal continuous, non-policy perturbation. Let dD_m and dS_m denote its direct demand and supply shifts at a fixed price. Hold every other primitive fixed. Assume continuously differentiable aggregates with finite derivatives locally and the derivative envelopes just stated. Supplier integration gives $S_{m,p} = \int 1/\Psi_{ss} dH_C > 0$, while demand monotonicity gives $D_{m,p} \leq 0$. Differentiating $Z(p_m) = D_m(p_m) - S_m(p_m) = 0$ at the unique equilibrium gives $S_{m,p} - D_{m,p} > 0$ and

$$dp_m^* = \frac{dD_m - dS_m}{S_{m,p} - D_{m,p}}. \quad (11)$$

A demand-increasing shifter raises the equilibrium price when supply is not shifted; a supply-increasing shifter lowers it when demand is not shifted. This formula is not applied to binary M .

Sufficient conditions, dimensions, and zero effects. The supplier signs use $\Psi_{ss} > 0$ and $\Psi_{sz} < 0$. The aggregate price derivative uses the additional local differentiability and derivative envelopes; strict monotonicity alone would not imply a nonzero slope. The derived positive supply derivative makes its denominator strictly positive. It is zero when the perturbation changes neither demand nor supply at the equilibrium. Equation (11) is dimensionally a change in CMO price.

Interpretation and prediction. These are CMO capacity and equilibrium-price margins. A higher capacity price induces more qualified supply but reduces study demand through both fewer entrusted assignments and weaker project advancement. The prediction is an upward-sloping supply response and a scarcity price that responds positively to continuous demand pressure, conditional on unchanged supply technology.

9.5 Binary reform at the equilibrium CMO price

Object and finite comparison. Let $p_m^h = p_m^*(h)$ for $h \in \{0, 1\}$. Supply technology, the supplier distribution, and background demand are held fixed. Since $D_m^{\text{MAH}}(p_m; 0) = 0$, market clearing at p_m^0 implies

$$Z_1(p_m^0) = D_m^{\text{MAH}}(p_m^0; 1) \geq 0.$$

Strict monotonicity of Z_1 therefore gives the finite comparison

$$p_m^1 \geq p_m^0, \quad p_m^1 > p_m^0 \iff D_m^{\text{MAH}}(p_m^0; 1) > 0. \quad (12)$$

No derivative with respect to M is used.

Define the fixed-price direct gain and the equilibrium gain as

$$\Delta\Omega_i^{\text{dir}} = \int [W_i^E(q, m; 1, p_m^0) - W_i^0(q, m)]_+ dF,$$

$$\Delta\Omega_i^{\text{eq}} = \int [W_i^E(q, m; 1, p_m^1) - W_i^0(q, m)]_+ dF.$$

Because $W_{i,p_m}^E = -b(m) < 0$, the price order yields

$$0 \leq \Delta\Omega_i^{\text{eq}} \leq \Delta\Omega_i^{\text{dir}}. \quad (13)$$

Applying the strictly increasing advancement mapping to these value gains gives

$$0 \leq \Delta x_i^{\text{eq}} \leq \Delta x_i^{\text{dir}}. \quad (14)$$

Conditions and zero effects. The price order requires policy-invariant supply/background demand and nonnegative post-reform study demand. Strictness requires positive study demand at p_m^0 . A developer's equilibrium gain is zero when the entrusted option remains below its old-route maximum at p_m^1 . Scarcity can eliminate an incremental gain but cannot lower the old choice-set value.

Economic interpretation and empirical prediction. This is an equilibrium CMO-price feedback on project advancement. New entrusted-route demand bids up scarce qualified capacity, so a fixed-price calculation overstates the equilibrium value and advancement response. The prediction is a weak post-reform increase in the CMO capacity price when the study cohort creates positive demand and an advancement response no larger than the fixed-price benchmark. A cross-market ranking by supply elasticity alone requires additional comparability conditions on demand and the pre-reform benchmark; it does not follow from this within-market comparison.

9.6 Planning-stage and realized-product finite comparisons

Objects and held-fixed primitives. The objects are planning-stage project intensity and realized retained-product outcomes, evaluated at the regime-specific equilibrium. Predetermined a_i , project distributions, realization probabilities, technologies and outside options are held fixed. The regime changes only route availability and the endogenous CMO price.

Planning-stage intensity satisfies the exact finite difference

$$\Delta\Lambda_i^{\text{plan}} = a_i\Delta x_i^{\text{eq}} \geq 0. \quad (15)$$

Let

$$Q_i^{\text{ret},h} = \int s(q)[\mathbf{1}\{r_i^{*,h} = I\} + \mathbf{1}\{r_i^{*,h} = E\}] dF.$$

Adding route E cannot remove an old option. If E wins, it is retained; if it does not win, the old maximizer is unchanged. Hence $Q_i^{\text{ret},1} \geq Q_i^{\text{ret},0}$. The exact decomposition is

$$\Delta Y_i^{\text{ret}} = a_i(x_i^1 - x_i^0)Q_i^{\text{ret},0} + a_i x_i^1(Q_i^{\text{ret},1} - Q_i^{\text{ret},0}) \geq 0. \quad (16)$$

The first term is project advancement; the second is retained-route composition. The same decomposition applies within class g using $\rho_g F_g$, $s_g = s$, and the common x_i , not a class-specific control. Zero conditional class surplus alone is not a zero-output condition when common advancement rises and that class retains positive realized mass.

Sufficient conditions and zero effects. A strict planning-stage increase requires a positive equilibrium expected-value gain. A strict retained-product increase requires a positive advancement term or positive-measure movement from T/A into E with positive realization weight. Reassignment from I to E raises observed holder–producer separation but, by itself, does not change the retained total. If advancement and route indicators are unchanged, every displayed outcome change is zero.

Economic interpretation and empirical prediction. These are planning-stage and realized-product margins. The model predicts that retained holder–producer separation can rise after route E becomes available, and separates that assignment change from the number of viable projects advanced and from downstream realization. Product-level holder and manufacturer records can test the assignment prediction. Patent applications are not an outcome in these expressions.

9.7 Mandatory limiting cases

Internal capability $k_i \rightarrow \infty$. On the feasible interior, $\Delta_{IE,k} > 0$, so increasing capability weakly favors I relative to E . Under the maintained upper-end crossing condition, $\Delta_{IE} > 0$ for sufficiently large k_i . This is a relative route-value statement; T or A may still dominate both.

Insufficient internal capability. If $k_i < \underline{k}(m)$, then $F_I(m, k_i) = +\infty$ and $W_i^I = -\infty$. Route I is infeasible rather than merely unattractive.

Pre-MAH regime. When $M = 0$, $\tau_E(0) = +\infty$, so $W_i^E = -\infty$, $\chi_i^E = 0$, and $D_m^{\text{MAH}} = 0$. The old route set and background CMO market remain.

Infinite CMO price. Because $b(m) > 0$,

$$\lim_{p_m \rightarrow \infty} W_i^E(q, m; 1, p_m) = -\infty.$$

For each fixed project, E eventually ceases to be optimal; no common finite exclusion price is asserted for unbounded project support. Thus

$$\lim_{p_m \rightarrow \infty} W_i(q, m; 1, p_m) = W_i^0(q, m). \quad (17)$$

The newly added option disappears economically but cannot reduce the old route-set value.

Zero entrusted-route advantage. If $W_i^E(q, m; 1, p_m) \leq W_i^0(q, m)$ for every draw, then

$$\Delta\Omega_i(p_m) = 0, \quad \Delta x_i(p_m) = 0. \quad (18)$$

At the equilibrium price, the same conclusion applies when the inequality holds at p_m^1 .

Perfectly elastic CMO supply. If qualified capacity is available at the common administered/competitive price $\bar{p}_m$ in both regimes, then

$$p_m^1 = p_m^0 = \bar{p}_m, \quad \Delta\Omega_i^{\text{eq}} = \Delta\Omega_i^{\text{dir}}, \quad \Delta x_i^{\text{eq}} = \Delta x_i^{\text{dir}}.$$

The equilibrium attenuation channel disappears. This is a boundary outside the strictly increasing-supply condition used for uniqueness, not a claim that baseline supply is perfectly elastic.

Quadratic advancement cost. For $\nu = 1$,

$$x_i^*(M, p_m) = \frac{\beta a_i}{\kappa} \Omega_i(M, p_m), \quad \Delta x_i = \frac{\beta a_i}{\kappa} \Delta\Omega_i. \quad (19)$$

The value response is exactly linear.

Novelty-share boundaries and no fixed ranking. The common-control decomposition is

$$\Delta\Omega_i = \rho_O \Delta\Omega_{iO} + \rho_{\text{Inc}} \Delta\Omega_{i\text{Inc}}. \quad (20)$$

If $\rho_O = 0$, the original-oriented class contributes zero to the aggregate; if $\rho_{\text{Inc}} = 0$, the incremental class contributes zero. Either conditional gain may be zero while the other is positive. Reversing the class supports of positive entrusted surplus reverses their ordering, so no fixed ranking follows from the baseline.

9.8 Comparative-static scope conclusion

All central signs above follow from the maintained primitive, technology, market, and mapping assumptions and the deterministic option-value structure. This section adds no route, control, state, market, welfare object, patent outcome, logit share, inclusive value or continuous implementation index. Its finite policy chain is

$$\begin{aligned} M &\longrightarrow \text{availability of retained entrusted manufacturing} \longrightarrow \Omega_i \\ &\longrightarrow x_i^* \longrightarrow \text{planning, route, and realized-product objects.} \end{aligned}$$

with the endogenous CMO price attenuating, rather than directly causing, the project-value response.